# Decision Modeling
## Week-4

## ⊛ M/D/1 Model :-

M → Markovian (Poisson)

D → Deterministic (Constant) ⟹ Service Rate

$L_q$ is $\frac{1}{2}$ of M/M/1 model system

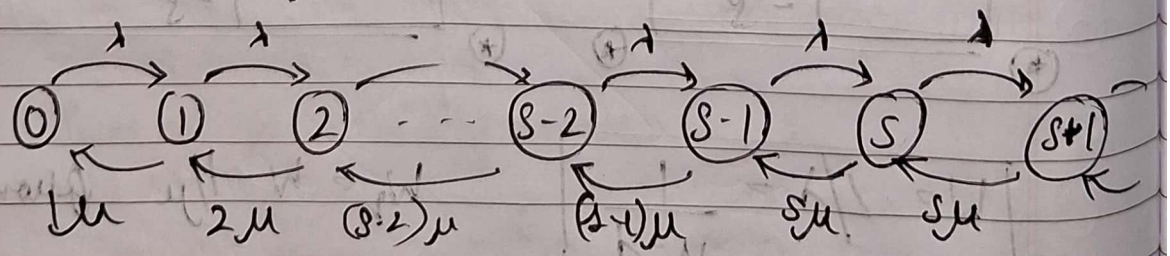
$$\boxed{L_q = \frac{1}{2}\frac{\rho^2}{(1-\rho)}}$$

$$\boxed{L = L_q + \rho}$$

Rest parameters by Little's law

## ⊛ M/M/s Model :-

In this model, no. of independent servers are s. They are all identical.




Steady State Condition :- $\boxed{\rho = \frac{\lambda}{S\mu} < 1}$

$$P_0 = \left[ \sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \cdot \frac{1}{1-(\lambda/s\mu)} \right]^{-1}$$

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} P_0 & \text{for } n = 1, 2, \ldots s \\[2ex] \dfrac{(\lambda/\mu)^n}{s!\, c^{n-s}} & \text{for } n = s+1, s+2 \ldots \end{cases}$$

# Utilization factor :- $\boxed{\rho = \dfrac{\lambda}{s\mu}} < 1$

# Expected no. of customers in queue ($L_q$) :-

$$L_q = \sum_{n=s}^{\infty} (n-s) P_n = \frac{(\lambda/\mu)^s \rho}{s!\,(1-\rho)^2} P_0$$

## Pooling of Service (Resources) :-

It always creates improvements in the system's performance.

Creates M/M/s from M/M/1

⇒ increases efficiency. ⇒ balanced utilization

⇒ shorter waiting time

## ⊛ M/M/∞ (Self-Service Model):

No. of identical servers are = ∞

**Steady State parameters:-**

Let $\boxed{a = \frac{\lambda}{\mu}}$ then,

$$\boxed{P_0 = e^{-a} = e^{-\lambda/\mu}}$$

$$\boxed{P_n = \frac{e^{-a} a^n}{a!}} \quad n = 0, 1, \ldots$$

$$\boxed{L = a = \frac{\lambda}{\mu}}$$

$$\boxed{L_q = 0}$$

$$\boxed{W = \frac{L}{\lambda} = \frac{1}{\mu}}$$

from Little's Law

$$W_q = \frac{L_q}{\lambda} = 0$$

# Queuing Cost Model

2 types of costs are involved in a queuing model

- Service Costs
- Waiting Costs

A trade-off is necessary b/w the 2 costs to obtain optimal service capacity

| Service Costs | Waiting Costs |
| --- | --- |
| ↓ | ↓ |
| Capital investment, cost of operation, cost of maintenance, repair, other costs | Cost of poor service, loss of customer relations, loss of repeat business due to dissatisfaction |

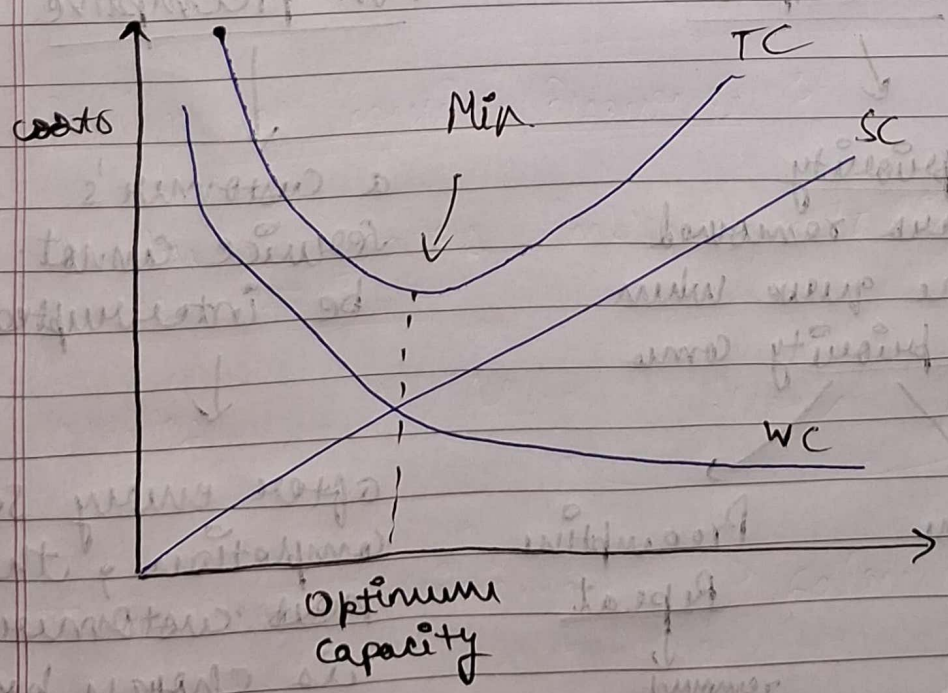
$$\boxed{\text{Total Queuing Cost (TC)} = \text{Service Cost (SC)} + \text{Waiting Cost (WC)}}$$

$C_W$ = Cost of waiting of 1 customer per unit time

$W * C_W$ = Avg cost of waiting per customer

For $\lambda$ customers per unit time,

$$\boxed{WC = \lambda * W * C_W}$$




TC = min for optimum value of = $\boxed{SC + WC}$

# Priority Discipline Queuing Models :-

| FCFS | SIRO | LCFS |
|---|---|---|
| First Come First Serve | Service In Random Order | Last Come First Serve |

$$\Rightarrow \boxed{E(W_{FCFS}) = E(W_{SIRO}) = E(W_{LCFS})}$$

However,

$$\boxed{Var(W_{FCFS}) < Var(W_{SIRO}) < Var(W_{LCFS})}$$

## Priority Queuing Models

- **Pre-Emptive**
  - Lower priority customers removed from the queue when higher priority comes
    - Preemptive Resume → removed customer service resumes
    - Preemptive Repeat → removed customer service repeats
- **Non-preemptive**
  - a customer's service cannot be interrupted
  - after every service completion, the next customer is chosen based upon the priority